



TEAM-02 · FCDF-COHORT-11 · LEVEL 1, SEMESTER 1

TCP/IP Communication

Tracing a Secure HTTPS Website Request Across the Network

Application Layer

Transport Layer

Internet Layer

Network Access Layer

Presented by

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2026/FCDF/16899

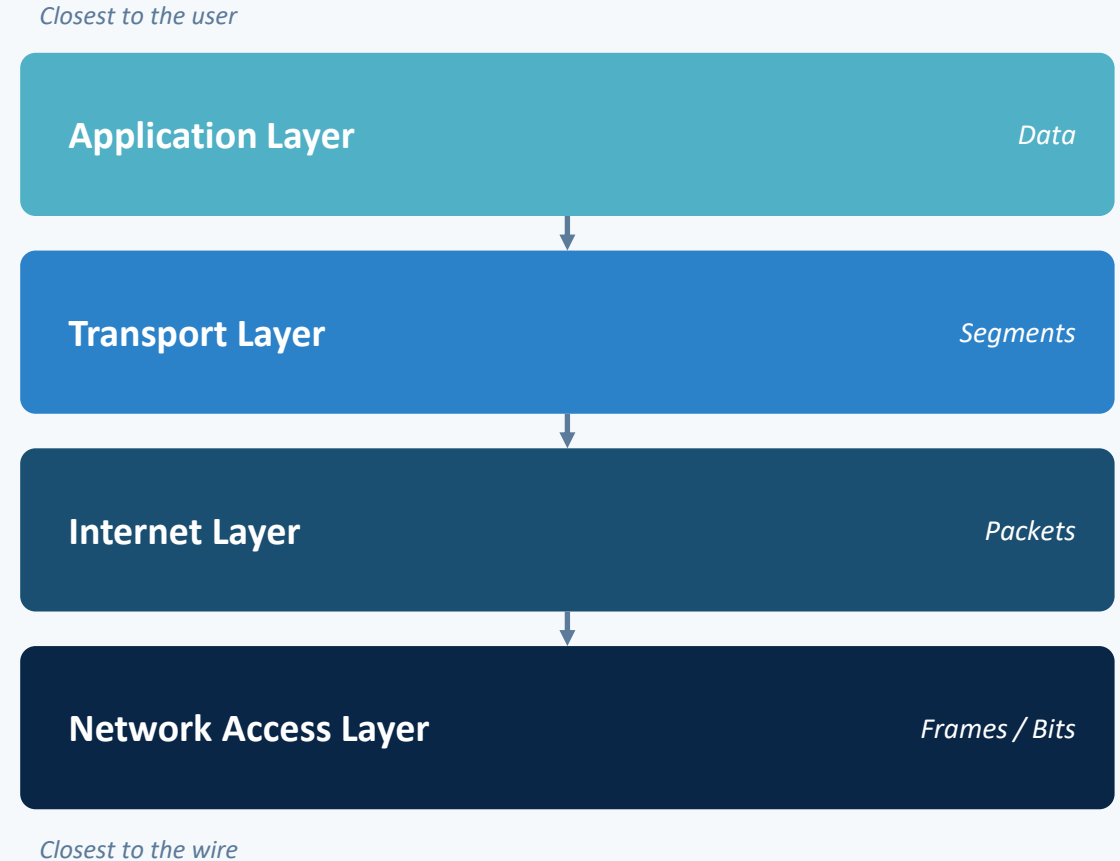
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Scenario: Opening a Secure Website (HTTPS)

What Is the TCP/IP Model?

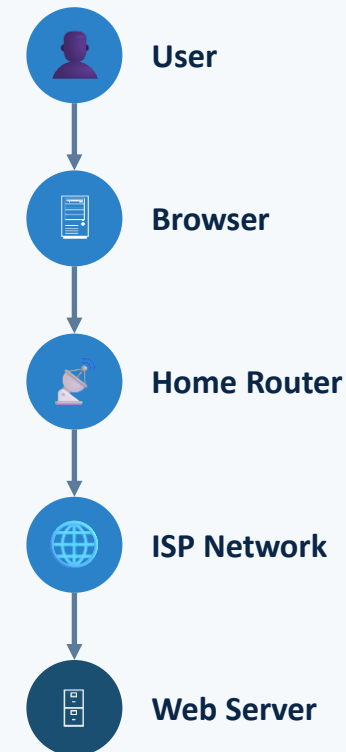
- TCP/IP (Transmission Control Protocol / Internet Protocol) is the rule set that governs how devices communicate over the Internet.
- It organizes communication into four layers, each with a distinct job — from generating a request to placing bits on the wire.
- Every online activity — browsing, email, video calls, file transfer — depends on all four layers working together.
- Our scenario (opening a secure website) exercises every layer, which is why it is a good vehicle for explaining the model.



Selected Scenario: Opening a Secure Website

User action: Typing `https://www.amazon.com` into a browser and loading the page over a secured HTTPS connection.

- The user's browser resolves the site's name and opens a secure (encrypted) connection to the web server.
- The request and response pass down through all four TCP/IP layers on the client and back up through all four on the server.
- This single request touches addressing (IP), delivery (TCP ports), routing (routers), and physical transmission (Wi-Fi/Ethernet) — making it ideal for showing the whole model end to end.
- We trace it step by step in the following slides: application → transport → internet → network access → across the Internet → and back.



Application Layer — Starting the Request

APPLICATION LAYER

- Function: generates the user-facing request and interprets the response — the only layer users directly interact with.
- Key protocols: HTTPS/HTTP (web pages), DNS (name resolution), TLS (encryption for HTTPS).
- DNS first resolves `www.amazon.com` to an IP address; then the browser builds an HTTPS request.
- Destination port: 443 (HTTPS) is the well-known port the server listens on for secure web traffic.

Sample request generated

```
GET / HTTP/1.1
Host: www.amazon.com
Destination Port: 443 (HTTPS)
```



Why HTTPS, not HTTP?

TLS encrypts the request before it ever leaves the Application Layer, protecting it from eavesdropping further down the stack.

Transport Layer — TCP Segmentation & Handshake

TRANSPORT LAYER

- Function: breaks application data into segments, adds port numbers, and guarantees reliable, ordered, error-checked delivery.
- Protocol used here: TCP (Transmission Control Protocol) — chosen because HTTPS needs guaranteed, in-order delivery.
- Port numbers: source = a random ephemeral port (e.g. 51322) on the client; destination = well-known port 443 on the server.
- Before any data moves, TCP opens the connection with a three-way handshake.



Once established, the client's HTTPS request is sent as one or more TCP segments.

Internet Layer — IP Addressing & Routing

INTERNET LAYER

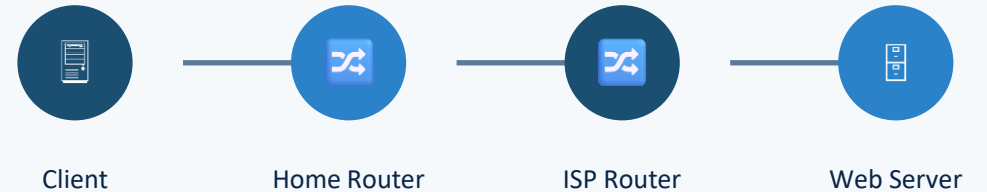
- Function: wraps each segment in an IP packet, adds source and destination IP addresses, and routes the packet toward the destination.
- Protocol used: IP (Internet Protocol) — acts like a postal address system for the network.
- Routers examine the destination IP address and forward the packet along the best available path, hop by hop.
- The same source/destination IP pair is used for both the request and the reply, so the server's response can find its way back.

Source IP (client):

192.168.1.10

Destination IP (server):

54.239.28.85

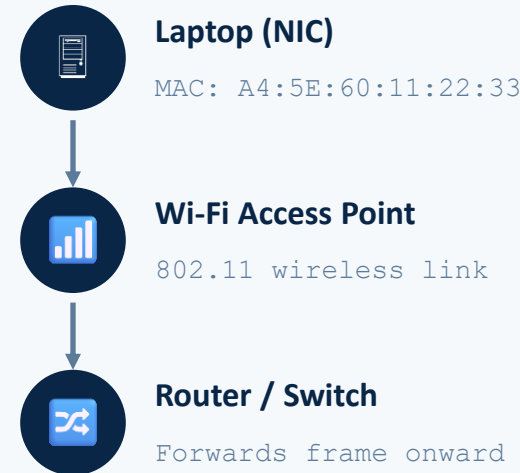


Packets are routed hop by hop using the destination IP address.

Network Access Layer — Physical Transmission

NETWORK ACCESS LAYER

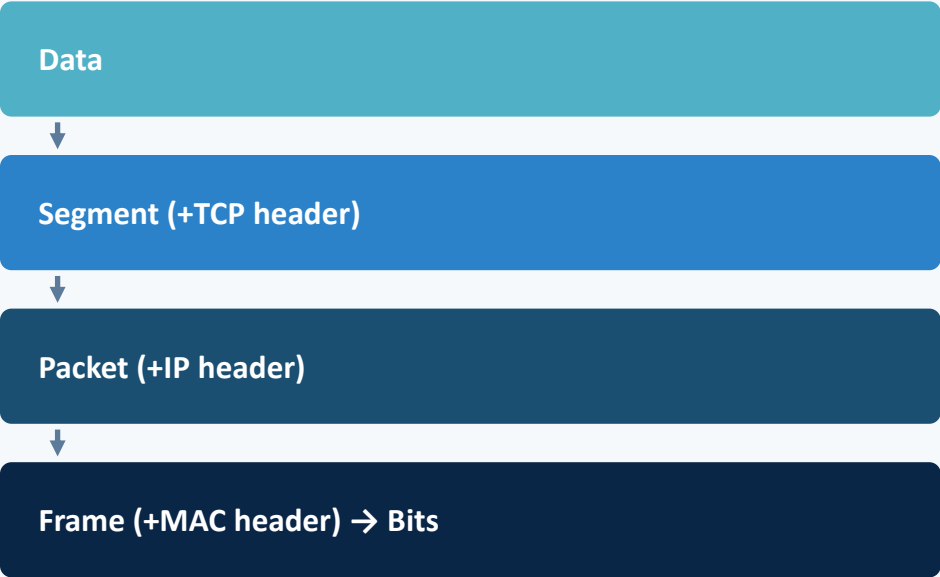
- Function: wraps the IP packet in a frame and physically transmits bits across a medium — Wi-Fi, Ethernet cable, or fibre.
- Uses MAC addresses — physical hardware addresses — to move the frame to the next device on the local network (e.g., the router).
- Network devices involved: network interface card (NIC), Wi-Fi access point / switch, and the router's local interface.
- This is the only layer that touches the physical medium — every layer above it is logical.



Encapsulation & Decapsulation

As data descends the client's stack, each layer adds its own header (encapsulation). The server reverses the process on receipt (decapsulation).

ENCAPSULATION (sender, top → bottom)



PROTOCOL ANALYSIS TABLE

Layer	Protocol	PDU	Addressing	Purpose
Application	HTTPS/DNS/TLS	Data	URL / domain name	Build & encrypt request
Transport	TCP	Segment	Port 443 / ephemeral	Reliable, ordered delivery
Internet	IP	Packet	Source / dest. IP	Logical addressing & routing
Network Access	Ethernet / Wi-Fi	Frame	Source / dest. MAC	Physical transmission

DECAPSULATION (receiver, bottom → top)

The web server receives bits, reassembles the frame, strips the MAC header, passes the packet up and strips the IP header, reassembles TCP segments and strips the TCP header, then hands the raw HTTPS request to the web application.

Mapping TCP/IP to the OSI Model

The 4-layer TCP/IP model is a practical, condensed version of the 7-layer OSI reference model.

TCP/IP Layer	Corresponding OSI Layer(s)	Example in Our Scenario
Application	Application, Presentation, Session	Browser builds HTTPS request; TLS session negotiated
Transport	Transport	TCP segments data; ports 443 / ephemeral used
Internet	Network	IP addresses attached; router forwards packet
Network Access	Data Link, Physical	Frame built with MAC address; bits sent over Wi-Fi

Note: OSI is a conceptual 7-layer reference model; TCP/IP is the practical 4-layer model actually implemented on the Internet.

TCP vs UDP — Choosing the Right Protocol

TCP

Transmission Control Protocol

Connection type

Connection-oriented (handshake first)

Reliability

Guaranteed delivery, with retransmission

Ordering

Data arrives in the order it was sent

Speed

Slower — overhead from reliability checks

Header size

20 bytes (larger)

Used by our scenario for

HTTPS web request (port 443)

Other examples

Email (SMTP), file transfer (FTP)

UDP

User Datagram Protocol

Connection type

Connectionless (no handshake)

Reliability

Best-effort — no delivery guarantee

Ordering

No guaranteed order

Speed

Faster — minimal overhead

Header size

8 bytes (smaller)

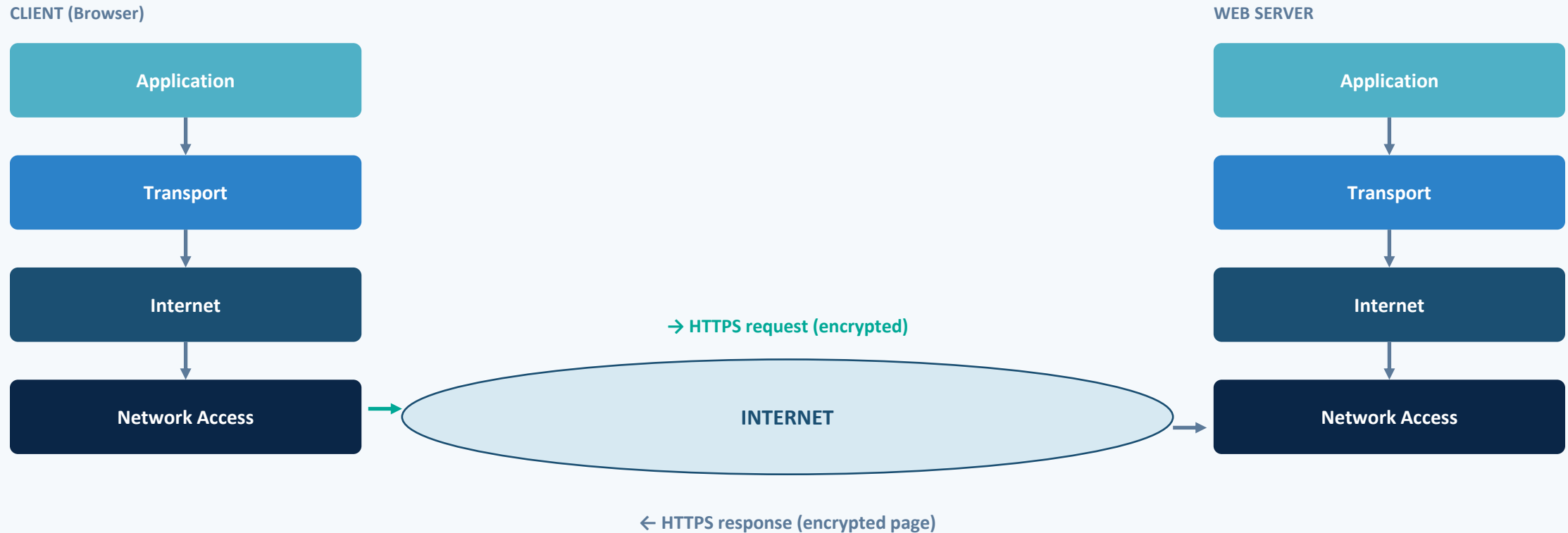
Not used here because

HTTPS needs guaranteed, ordered delivery

Typical examples

Video streaming, VoIP calls, online gaming, DNS queries

Complete Packet Journey — Request & Response



Data travels down the client's stack, across the Internet, and up the server's stack for the request — then the reverse path returns the response.

Troubleshooting a Communication Failure

Failure scenario: The user's browser shows “This site can’t be reached” when opening <https://www.amazon.com>.

1

Confirm physical connectivity

Network Access

Check Wi-Fi/Ethernet link light and adapter status; retry on another network if needed.

2

Verify IP configuration

Internet

Run `ipconfig / ifconfig` and ping the default gateway to confirm the device has a valid IP.

3

Test DNS resolution

Application

Run `nslookup www.amazon.com` — failure here means the name isn’t resolving to an IP.

4

Check port & firewall

Transport

Confirm port 443 isn’t blocked by a local firewall or the ISP; test with `telnet/Test-NetConnection`.

5

Inspect the TCP handshake

Transport

Capture traffic (e.g., Wireshark) to see if SYN/SYN-ACK/ACK completes; an incomplete handshake points to a network path issue.

6

Validate the TLS certificate

Application

If the handshake completes but HTTPS fails, check for an expired/invalid certificate on the server.

CONCLUSION

Key Takeaways

- TCP/IP's four layers each add a specific header, enabling reliable end-to-end communication.
- Ports (443/ephemeral) and IP/MAC addresses together get data to the right application on the right device.
- TCP trades speed for reliability — the right choice for HTTPS; UDP trades reliability for speed — right for streaming, VoIP, gaming and DNS.
- The OSI model maps cleanly onto TCP/IP and gives a more detailed vocabulary for the same process.
- A layer-by-layer troubleshooting approach (physical → IP → DNS → port → handshake → TLS) isolates failures quickly.

Application

Transport

Internet

Network Access

Thank you — Questions?

Team Contribution Record

Member	Slides Presented	Contribution
Maryjudith Chidinma Ogunaka C11/26/FCDF/17151	Slides 1 - 7	Presented the title, introduction to the TCP/IP model, the selected scenario, and the Application, Transport, Internet and Network Access layers.
Abdul Fatawu Umar 2026/FCDF/16899	Slides 8 - 13	Presented encapsulation & decapsulation, the OSI mapping, TCP vs UDP comparison, the packet journey diagram, troubleshooting, and the conclusion.

References

1

RFC 9293 — Transmission Control Protocol (TCP)

IETF · Official Internet standard defining TCP.

www.rfc-editor.org/rfc/rfc9293

2

Introduction to Networks — The OSI and TCP/IP Models

Cisco Networking Academy · Recognised vendor networking-academy course material.

www.netacad.com

3

What Happens When You Type a URL Into Your Browser?

Cloudflare Learning Center · Vendor explainer on the request/response journey.

www.cloudflare.com/learning

4

An Overview of HTTP

MDN Web Docs (Mozilla) · Reference documentation on HTTP/HTTPS.

developer.mozilla.org